

Factor of Safety (Machine Design)

https://youtu.be/EoN0_TTv8Mk



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Factor of Safety

Factor of Safety (FoS) is how much could a system withstand beyond the expected loads or actual loads. It's a kind of reserve strength to avoid accidents and failures.

- **Permissible stress:** The greatest stress which a designer expects for a machine component or machine. It is used to design the component or the machine.
- Permissible stress is also termed as : **allowable stress or working stress or design stress**

Design steps:

1. Material selection: Ultimate or Yield strength is available
2. Assume suitable FOS : considering various factors
3. Calculate Permissible stress:

Permissible stress = Ultimate strength/ Factor of safety (For Brittle materials)

or

Permissible stress = Yield strength / Factor of safety (For Ductile materials)

Factor of safety depends upon:

- **Degree of safety required:** If safety required is more such as in human hazards or high financial loss possibility, factor of safety will be higher.
For example: Areas where human hazards are involved we would require more safety like boiler applications, Bridges etc.
- **Nature of loading:** The factor of safety varies greatly according to the nature of the loads,
For example: static, variable or alternating, impact loadings.
- **Accuracy of problem formulations:** If the estimation of stresses and load is not accurate we have to take higher factor of safety.
For example: Lower FOS is required in case of precisely determined loads. Higher FoS is chosen for Design of a Bus as compared to an Elevator.
- **Degree of homogeneity :** The factor of safety will be lower if material used is of uniform properties.
For example: Ductile materials like steel would require lower FOS as compared to Cast iron. Also it depends upon the quality of raw materials supplied by the vendor.

Factor of safety depends upon:

- **Expected reliability:** As the expected reliability increases, FOS increases.

For example the components which are going to be used in power stations or defense applications would require higher factor of safety

- **Testing Accuracy:** Testing conditions affect the value of FOS chosen.

For example: FOS will be higher for equipment which can not be tested in actual running conditions

- **Service conditions:** Higher FOS is to be chosen for the components which are going to operate in corrosive or high temperature environment.
- **Manufacturing processes:** If the quality of manufacturing methods used is good factor of safety required is lesser.
- **Experience of Design engineer:** Newer component needs higher FoS as compared to the repeated product.
- **Degree of economy desired:** If we go for higher factor of safety the cost of the component will be more as costs of material, manufacturing and testing are increased.

Factor of safety depends upon the uncertainty and fear of loss

Problem 1.2 A hollow steel column of external diameter 250 mm has to support an axial load of 4000 kN. If the ultimate stress for the steel column is 480 N/mm². Find the internal diameter of the column allowing a factor of safety of 4.

Solution: External diameter $D = 250 \text{ mm}$

Internal diameter $d \text{ mm}$

Load on the column $P = 4000 \times 1000 \text{ N}$

Ultimate stress $= 480 \text{ N/mm}^2$

Factor of safety $= 4$

$$\therefore \text{ Safe stress } \sigma = \frac{480}{4} = 120 \text{ N/mm}^2$$

$$\therefore \text{ Sectional area required } A = \frac{P}{\sigma} = \frac{4000 \times 1000}{120} = 33333.33 \text{ mm}^2$$

$$\therefore \frac{\pi}{4}(250^2 - d^2) = 33333.33$$

$$\therefore 62500 - d^2 = 21221$$

$$\therefore d^2 = 29167$$

$$\therefore d = 170 \text{ mm}$$